上課注意事項:

1. 圖書館內禁止攜帶食物飲料入內。
2. 請實體與線上學員掃描以下QR code進行線上簽到。
3. 請線上學員於課程開始前關閉自己的麥克風。
4. 線上學員若有問題, 請先按下“舉手”, 或於聊天室寫下問題, 將於課程結束後在場地時間允許下, 安排QA時間。
5. 現場學員發問時請使用麥克風才可進行收音。
6. 台大的線上學員請優先使用非台大的網路參與課程, 以確保連線品質。

線上簽到

課程材料與資訊連結

Napari-Zelda 傳說：
解鎖 3D 影像分割及物體間關聯分析

3D segmentation and parent-child relation analysis with napari-zelda

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Post-Doctoral Researcher, Chin-Hsien Lin Lab, Department of Neurology

National Taiwan University Hospital, Taipei, Taiwan

2026.04.13

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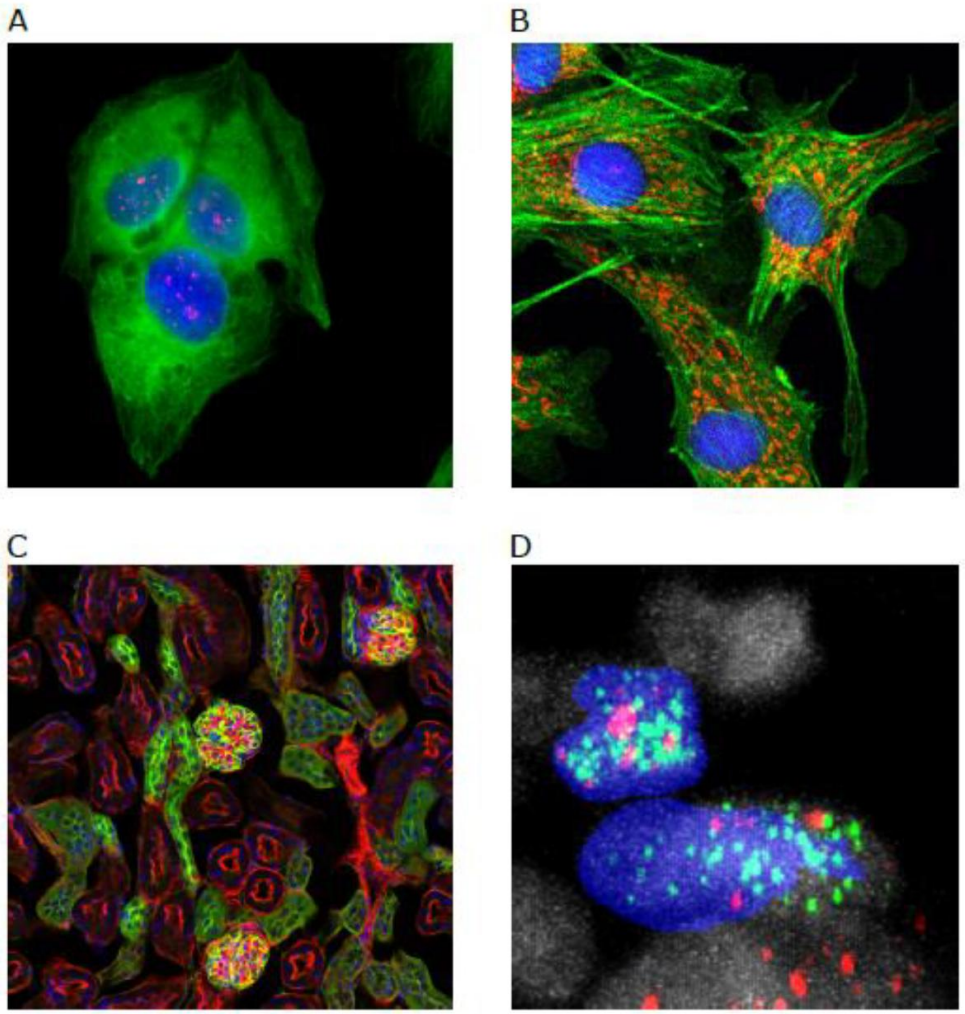
Some common bioimage analysis problem

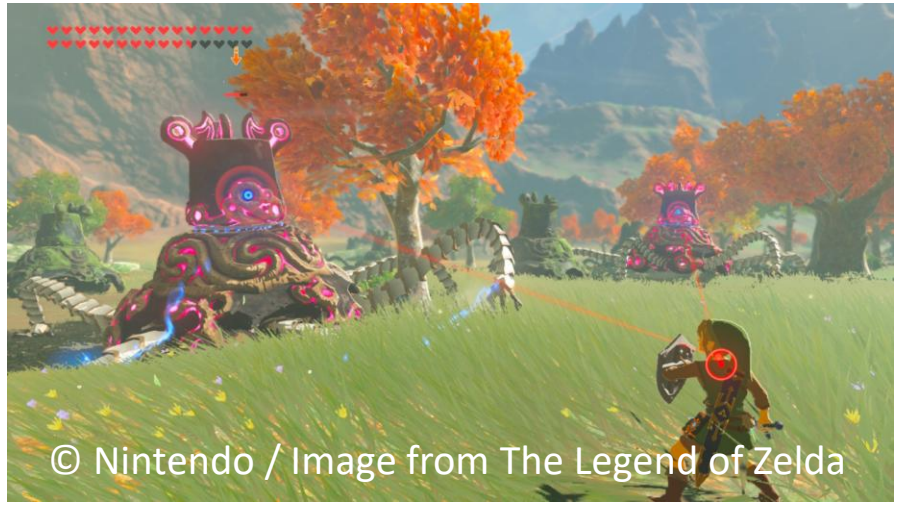


2D or 3D images:

“segmentation”,

“counting”,

and **“parent-child relation”**
物體關聯分析



‘Point and click’ segmentation and parent-child relation analysis workflow

ZELDA: a 3D Image Segmentation and Parent-Child relation plugin for microscopy image analysis in napari

https://github.com/RoccoDAnt/napari-zelda

D’Antuono, Rocco, and Giuseppina Pisignano. 2022. “ZELDA: A 3D Image Segmentation and Parent-Child Relation Plugin for Microscopy Image Analysis in Napari.” *Frontiers in Computer Science*3: 115. https://doi.org/10.3389/fcomp.2021.796117

Video tutorial: "Napari Zelda Tutorial on 3D image segmentation for microscopy - Rocco D'Antuono" - LIBRE Hub YouTube channel

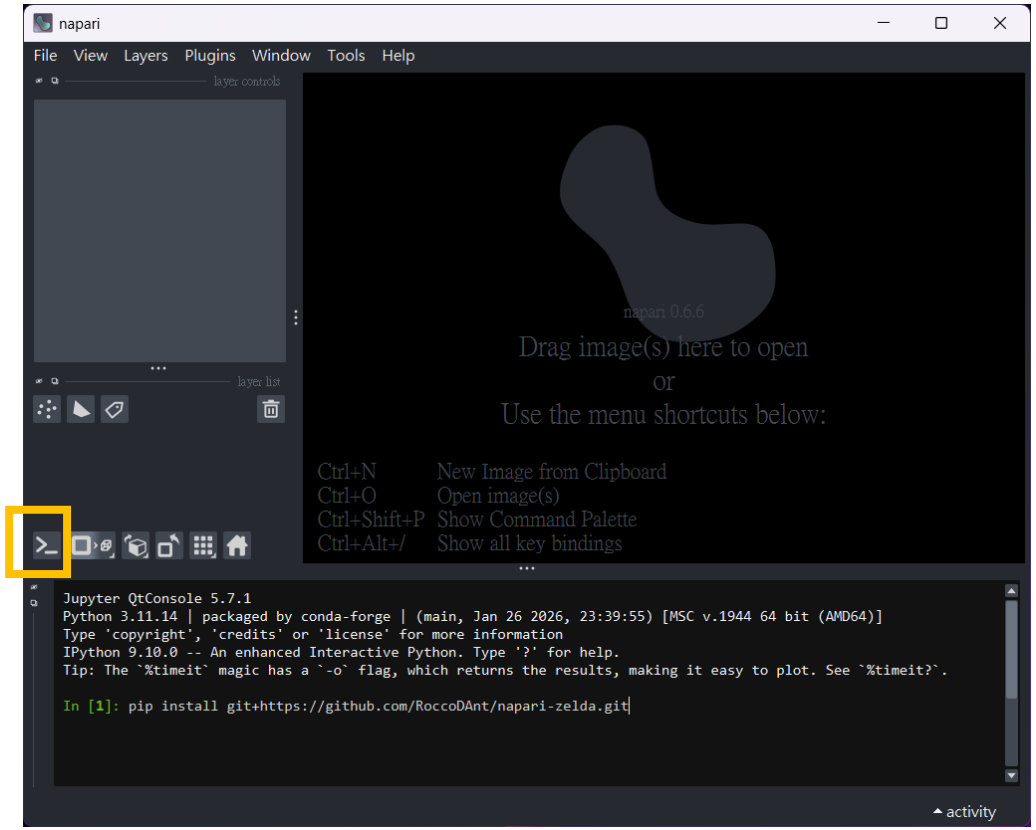
Course outline

- Napari-zelda GUI
  - 利用zelda pipeline 進行大物件與小物件之間 parents-children relation analysis
  - zelda的數據繪圖與建立protocol/pipeline功能介紹
- Jupyter notebook 與 napari-zelda
  - 使用Jupyter notebook 控制 napari-zelda
- Jupyter notebook batch 執行 parents-children relation analysis
  - 使用Jupyter notebook ，達到batch執行parents-children relation analysis

安裝 napari-zelda (需要連網路)

- 開啟 napari GUI
- 開啟 (integrated python console)
- 輸入 pip install git+https://github.com/RoccoDAnt/napari-zelda.git

- 關閉並重新啟動 napari GUI
- Plugins > launch_ZELDA (napari-zelda)



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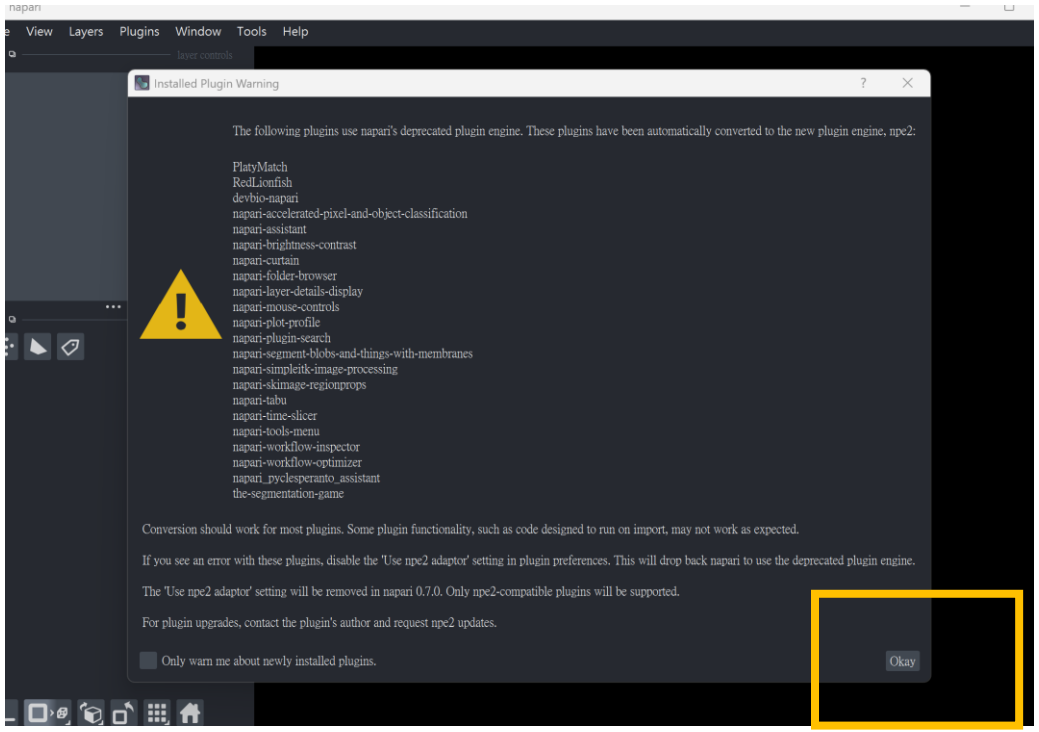
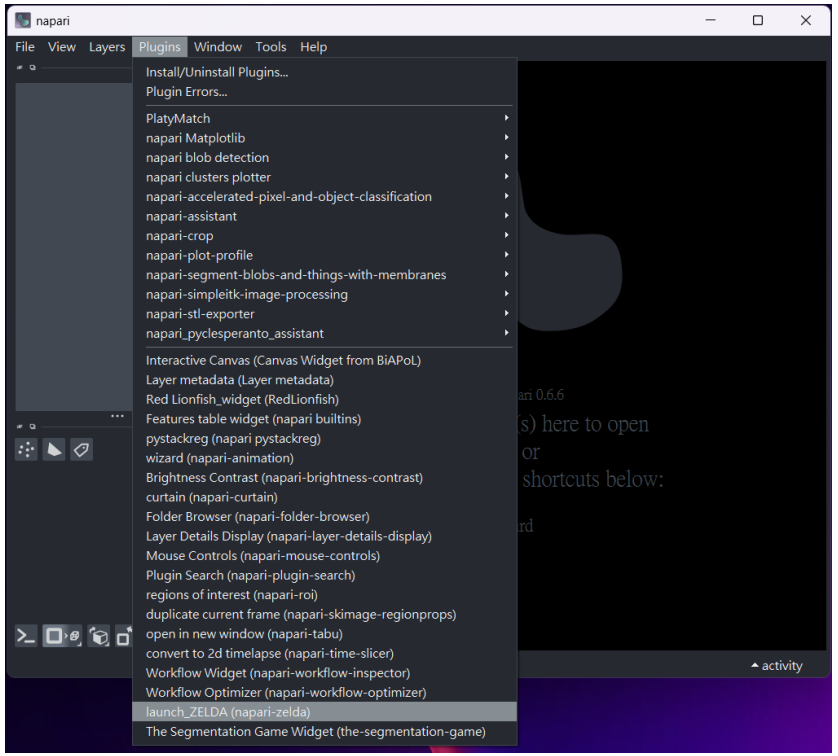
  - 使用Jupyter notebook ，達到batch 執行parents-children relation analysis

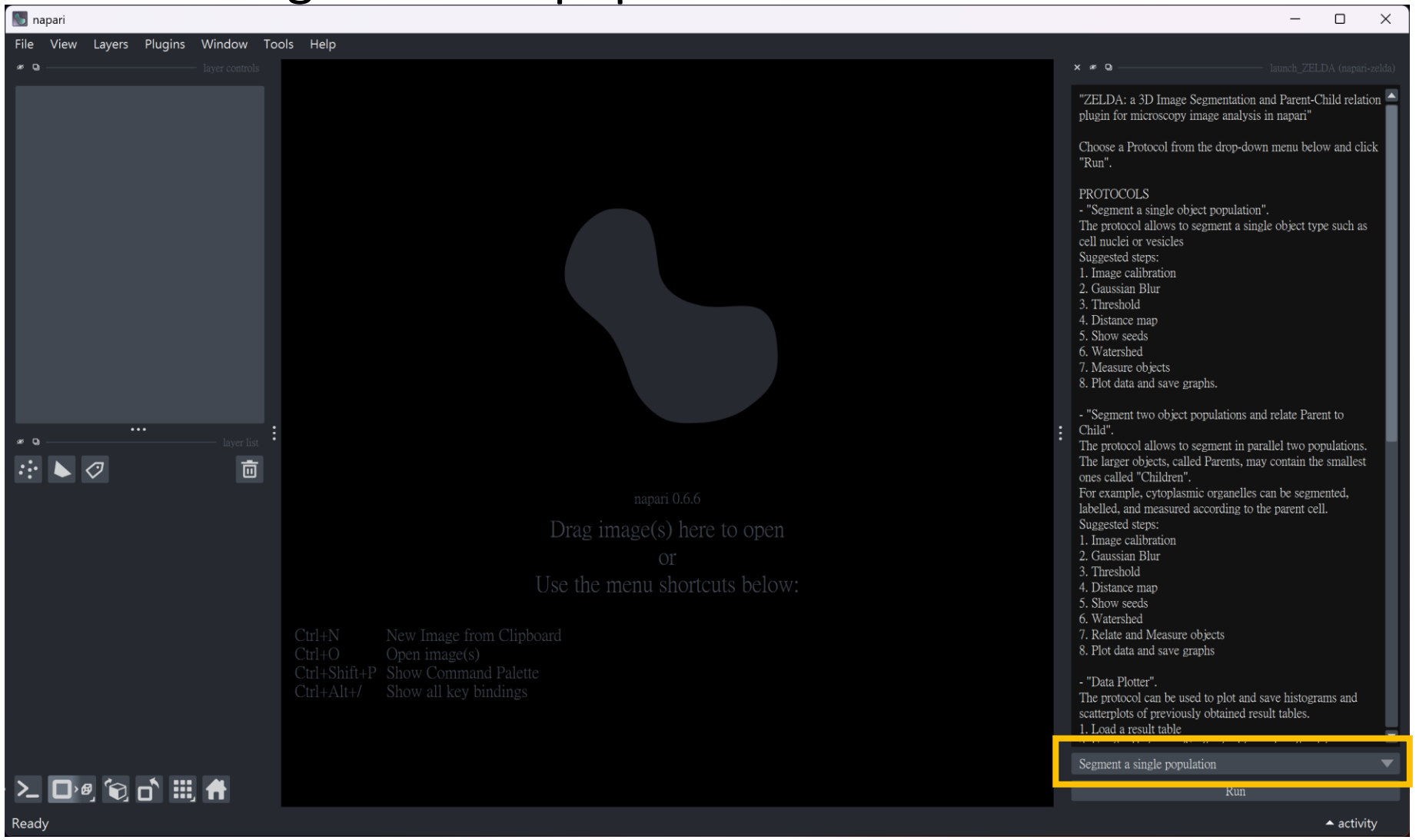
Launch napari-zelda

1. Open prompt activate 你的napari環境
2. (base) C:\Users\User\>conda activate EBIAS_TP_2026
3. (EBIAS_TP_2026) C:\Users\User\>napari
   GUI出現提示按Okay
4. Plugins → launch_ZELDA (napari-zelda)





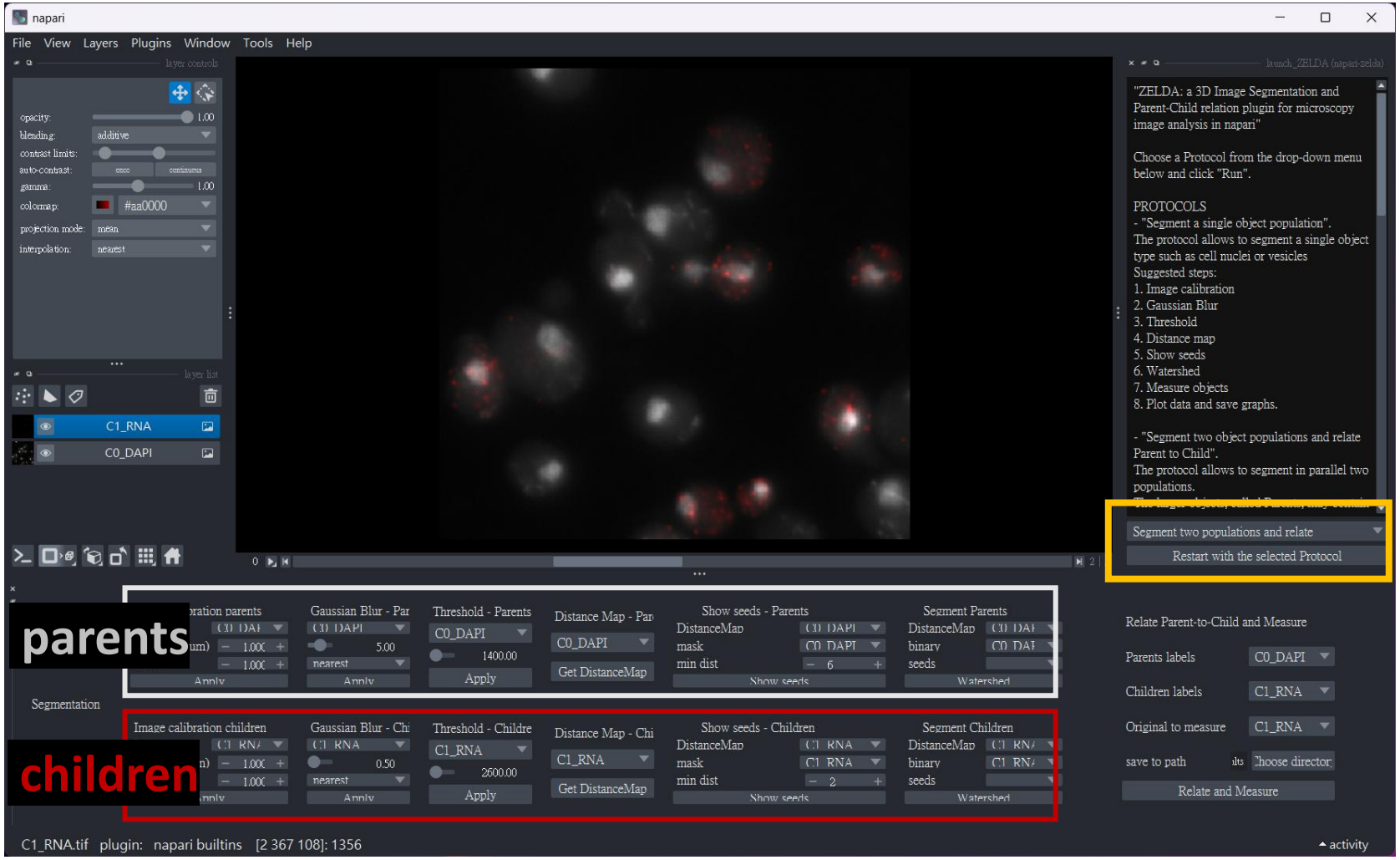
Launch napari-zelda
→ Segment two populations and relate → Run



Segment two populations and relate

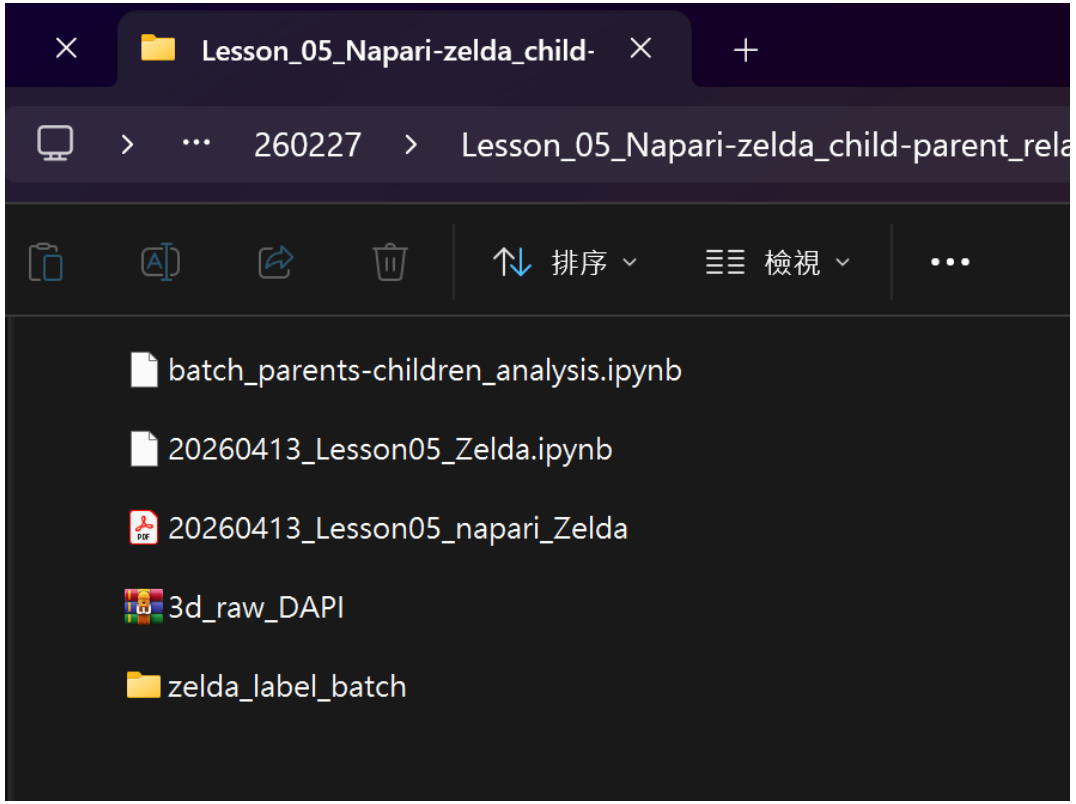
2 segmentation work flows (for parents and for children)

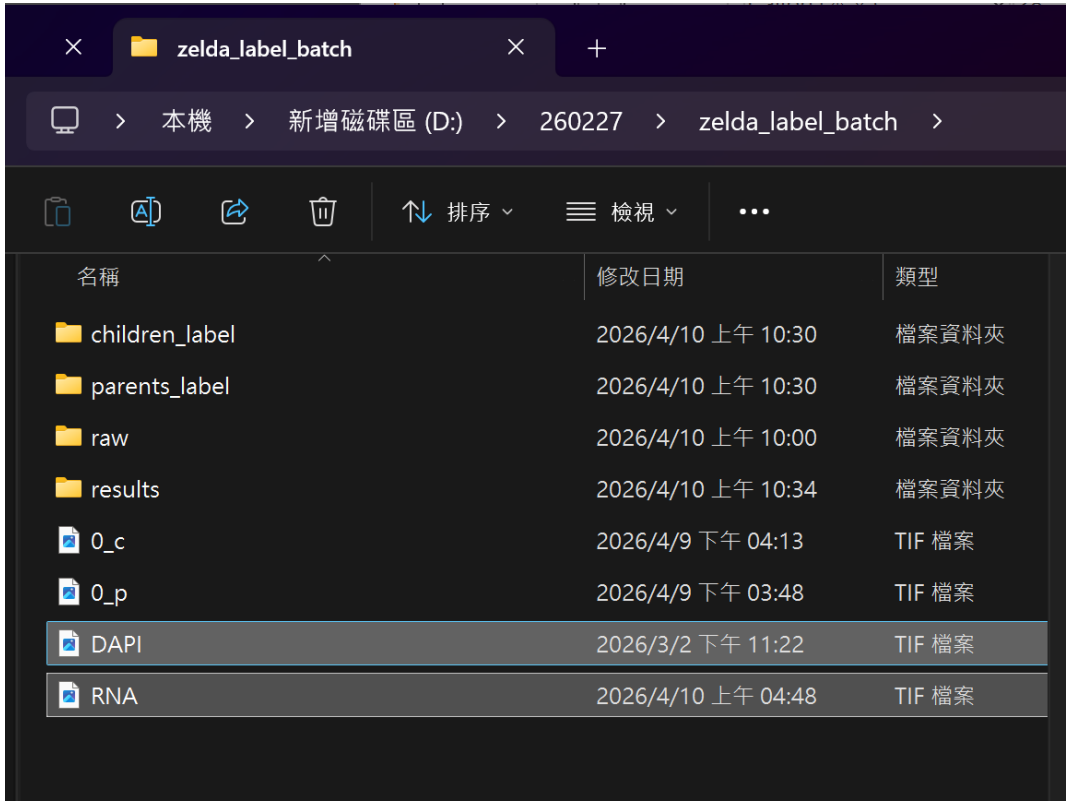
and 1 parent-child relation



https://idr.openmicroscopy.org/webclient/?show=image-4496770

Open sample image
zelda_labe_batch → DAPI.tif, RNA.tif





Segment two populations and relate

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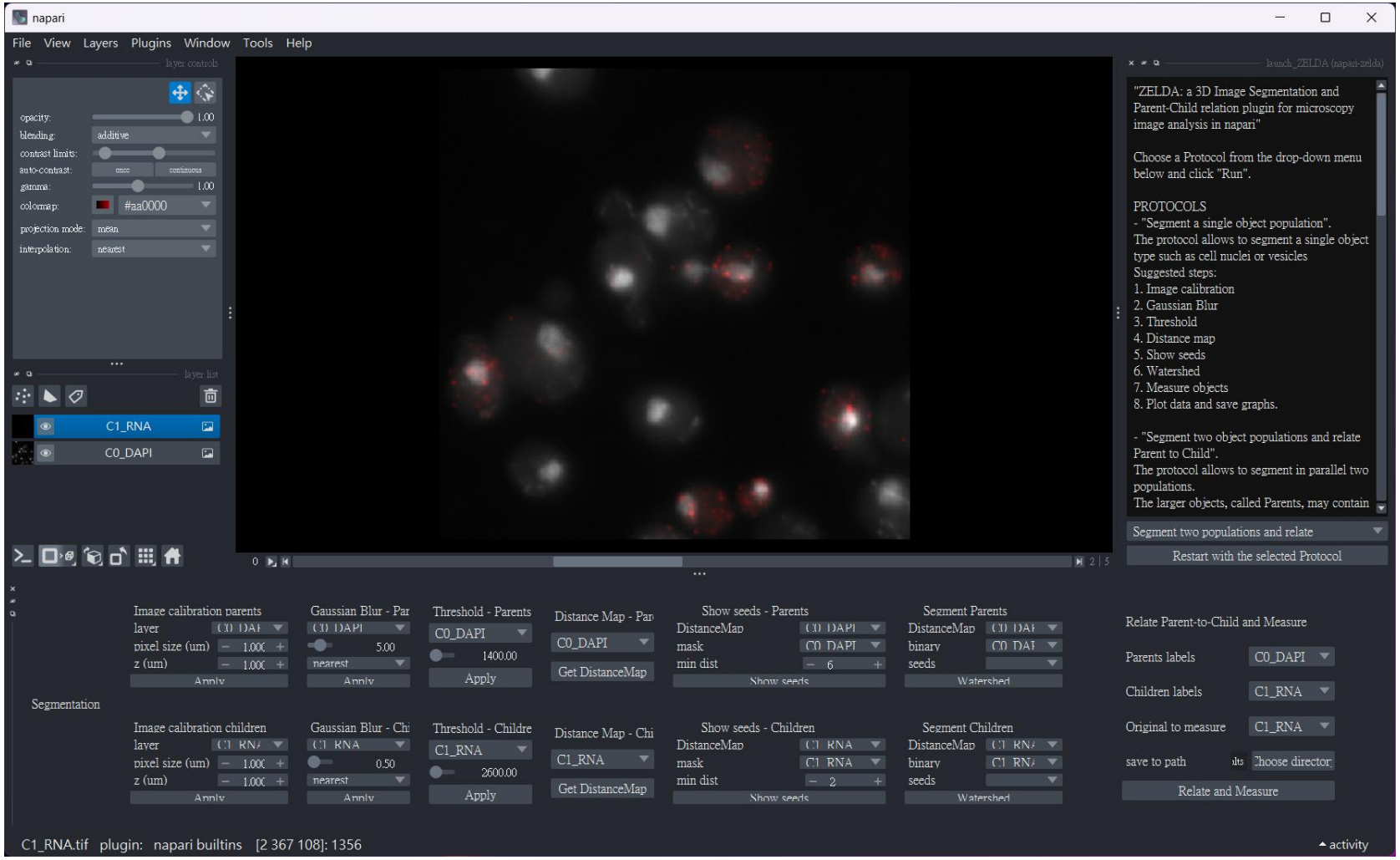
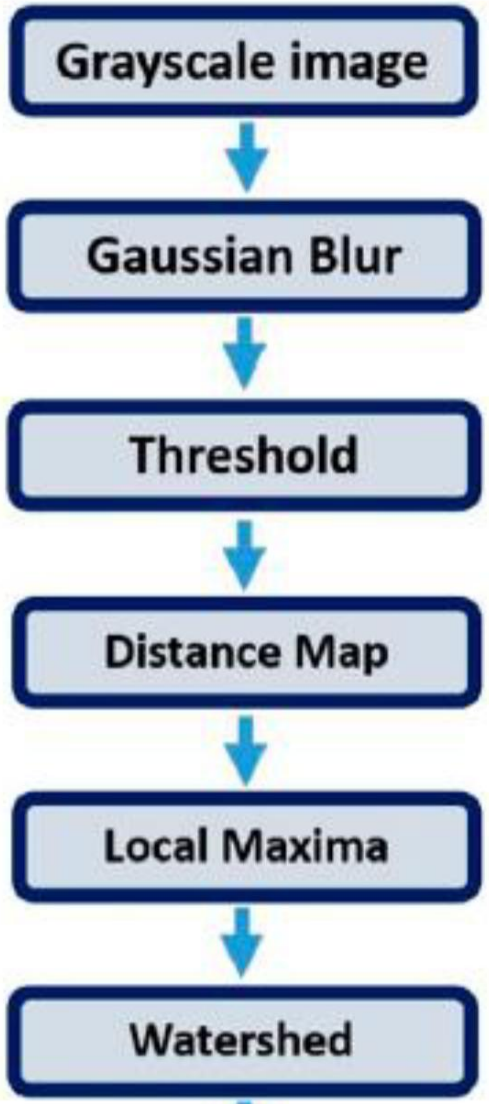
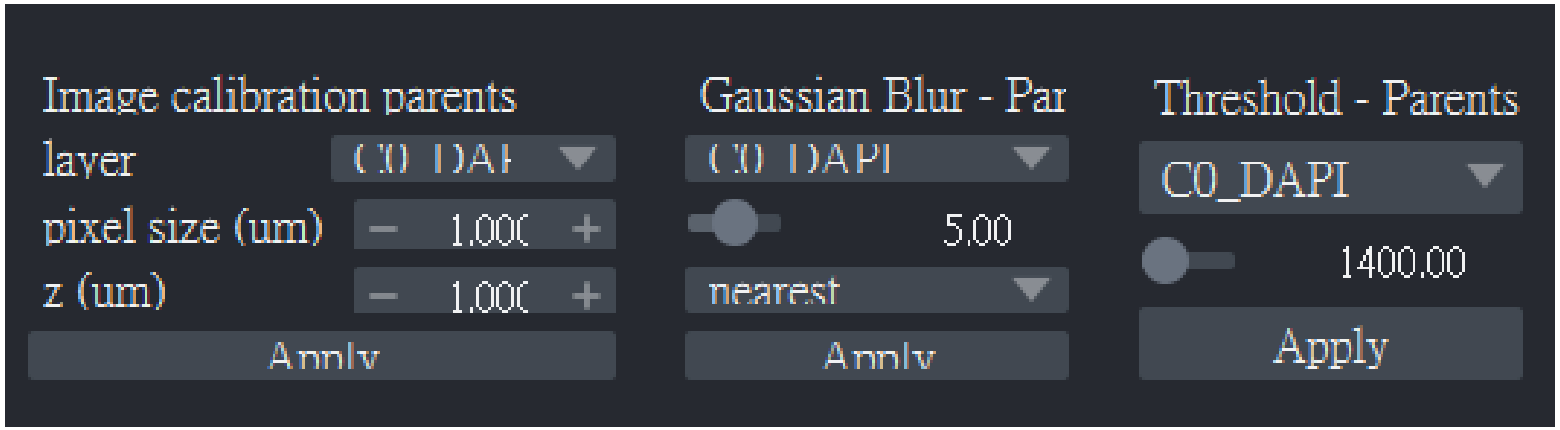
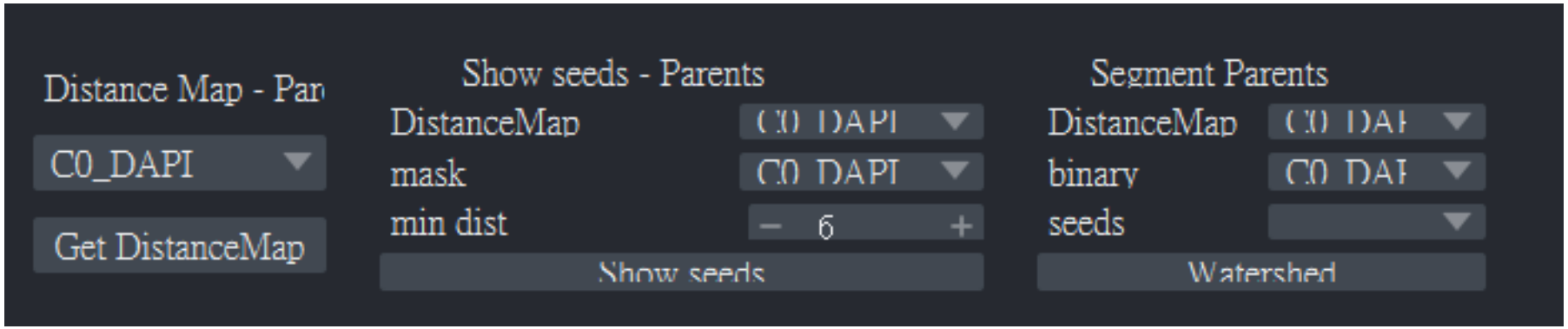


https://idr.openmicroscopy.org/webclient/?show=image-4496770

Segmentation for a single population of objects

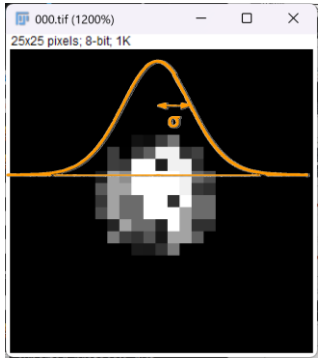


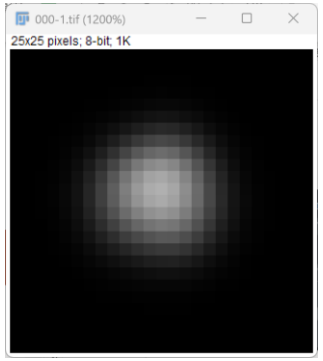




Gaussian Blur - Threshold - Distance Map – Seeds - Watershed

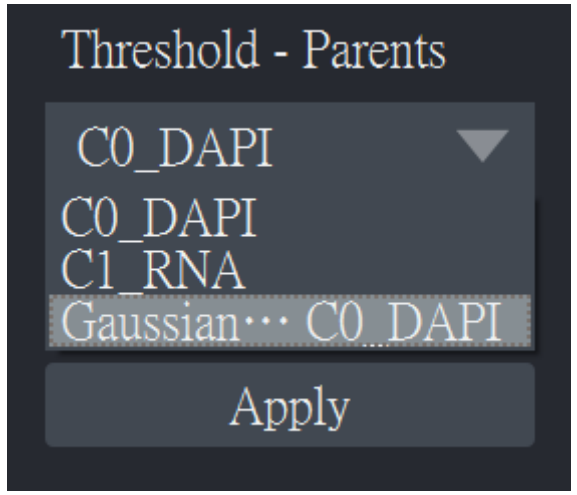
- Gaussian blur:
  - sigma 約物體直徑 1/2 (1/3) (單位 pixels)



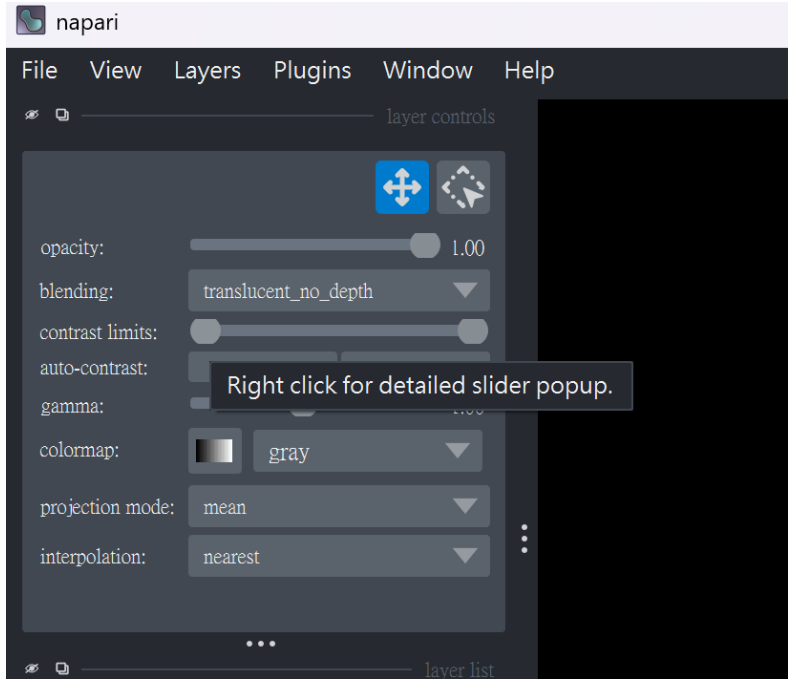


  - methods:
    - nearest: [a, a | a, b, c, d | d, d]
    - constant: [0, 0 | a, b, c, d | 0, 0]
    - reflect: [d, c, b | a, b, c, d | c, b, a]
    - mirror: [c, d | a, b, c, d | a, b]
    - wrap: [c, d | a, b, c, d | a, b]

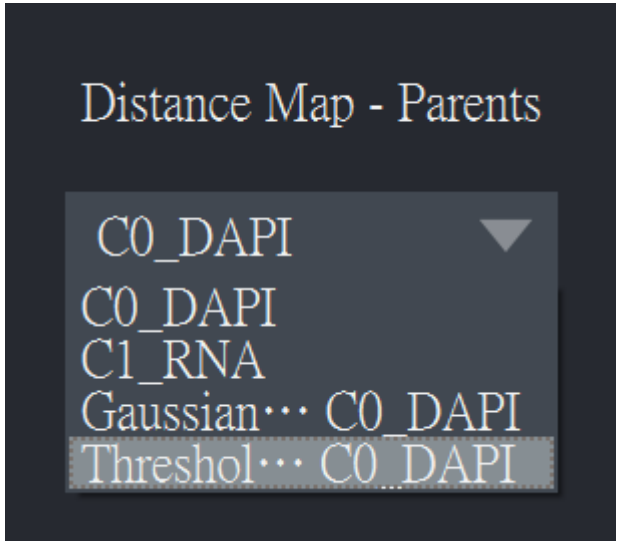
Gaussian Blur - Threshold - Distance Map – Seeds - Watershed



- Threshold：前景為1，背景為0的臨界值
  - ➢利用 Layer control → contrast limits 滑鼠右鍵輔助判斷

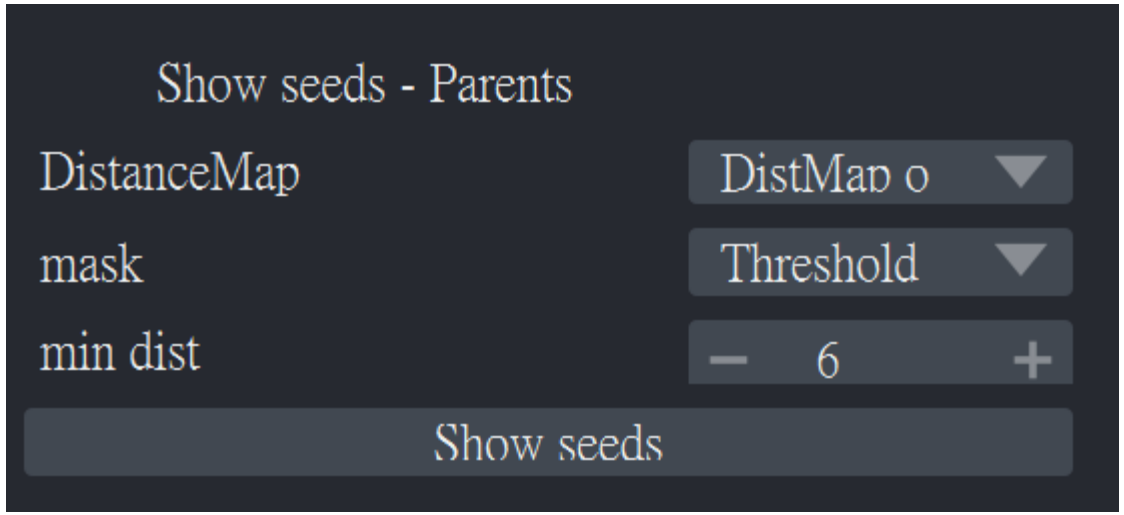


Gaussian Blur - Threshold - Distance Map – Seeds - Watershed

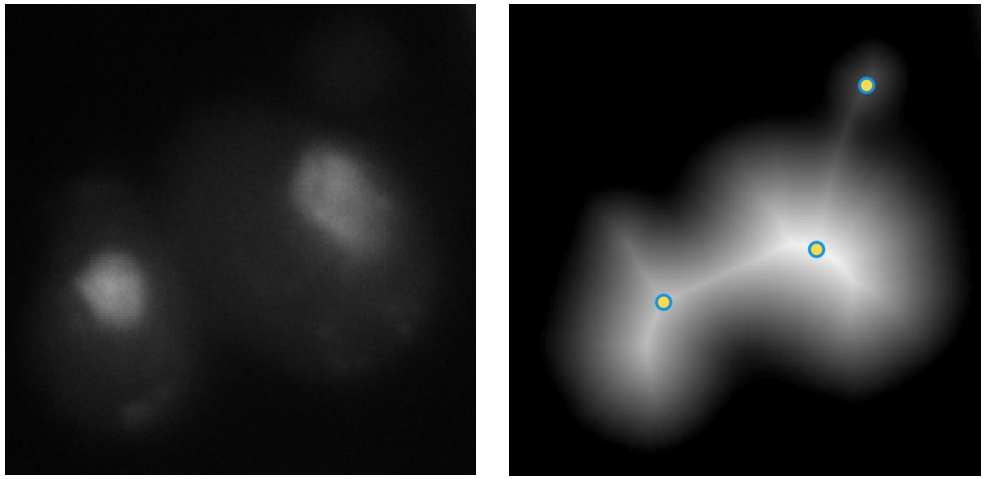


- Distance Map
  - 計算 Threshold mask 中，前景pixel 到最近背景pixel 的距離

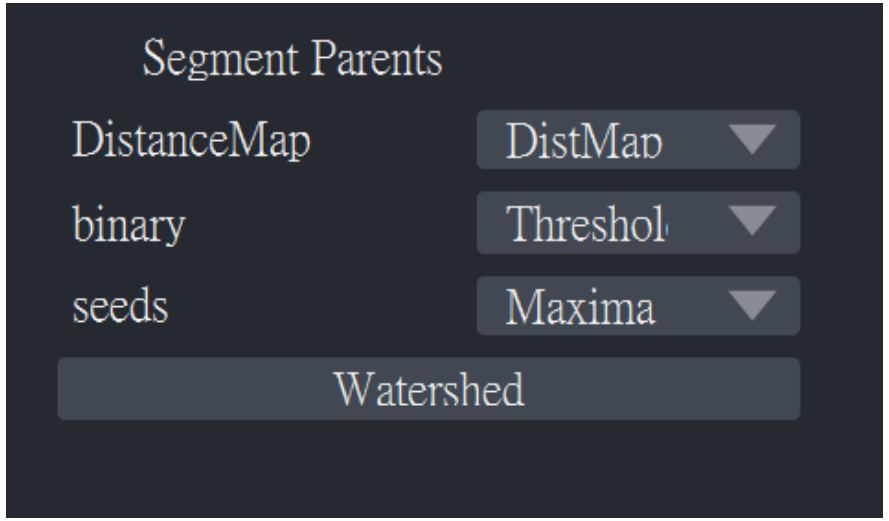
Gaussian Blur - Threshold - Distance Map – Seeds - Watershed



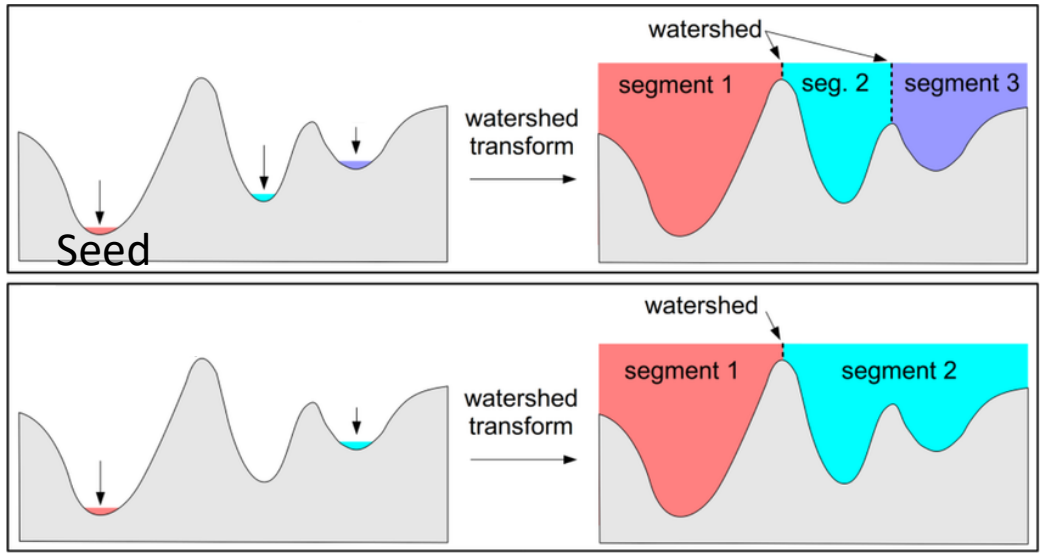
- Seeds :
  - ➢min dist : 距離多遠的local maximum



Gaussian Blur - Threshold - Distance Map – Seeds - Watershed



- Watershed :
  - ➢DistanceMap : DistMap
  - ➢binary : Threshold
  - ➢seeds : Maxima

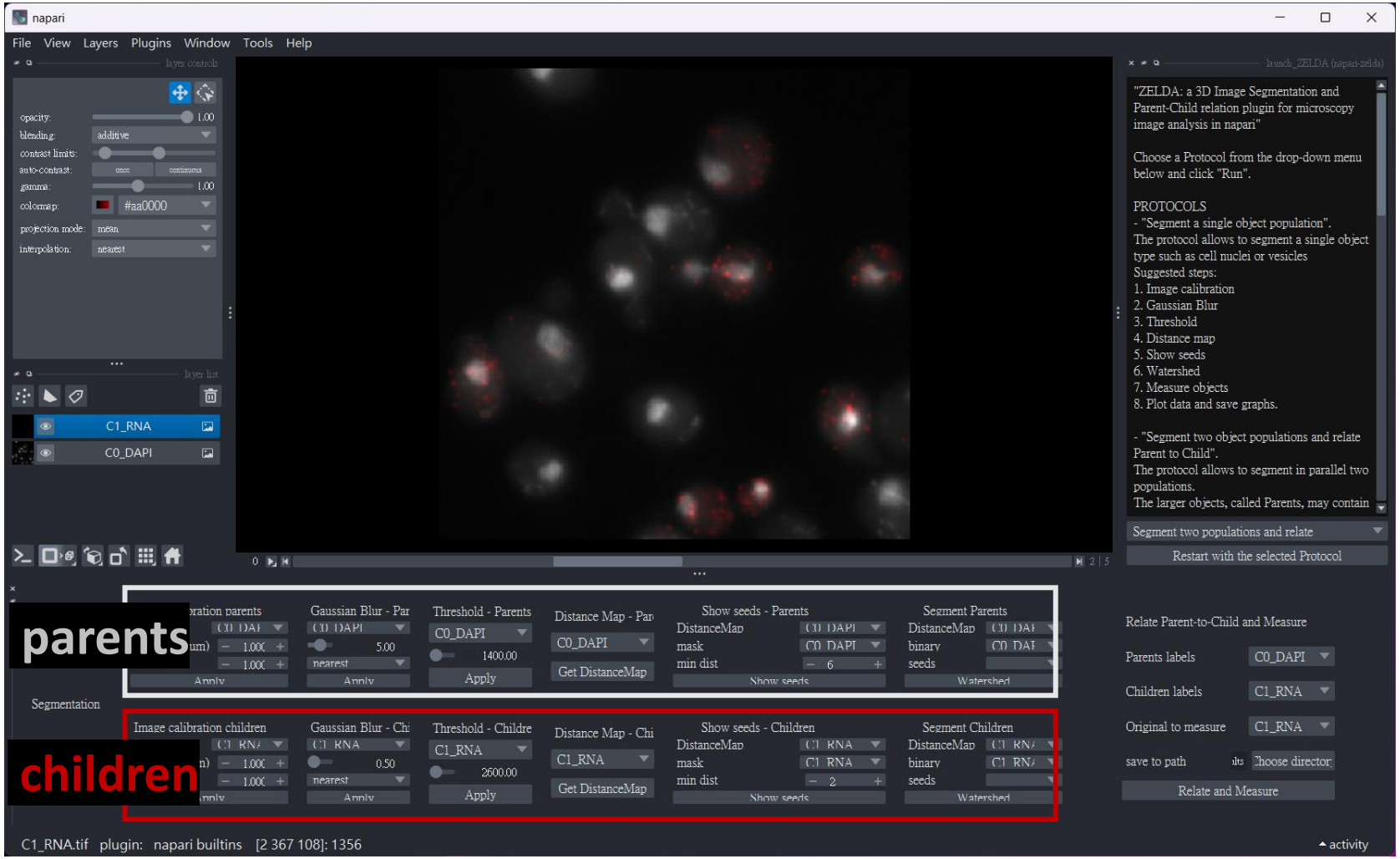


DOI:10.13140/RG.2.2.35718.98889

Segment two populations and relate

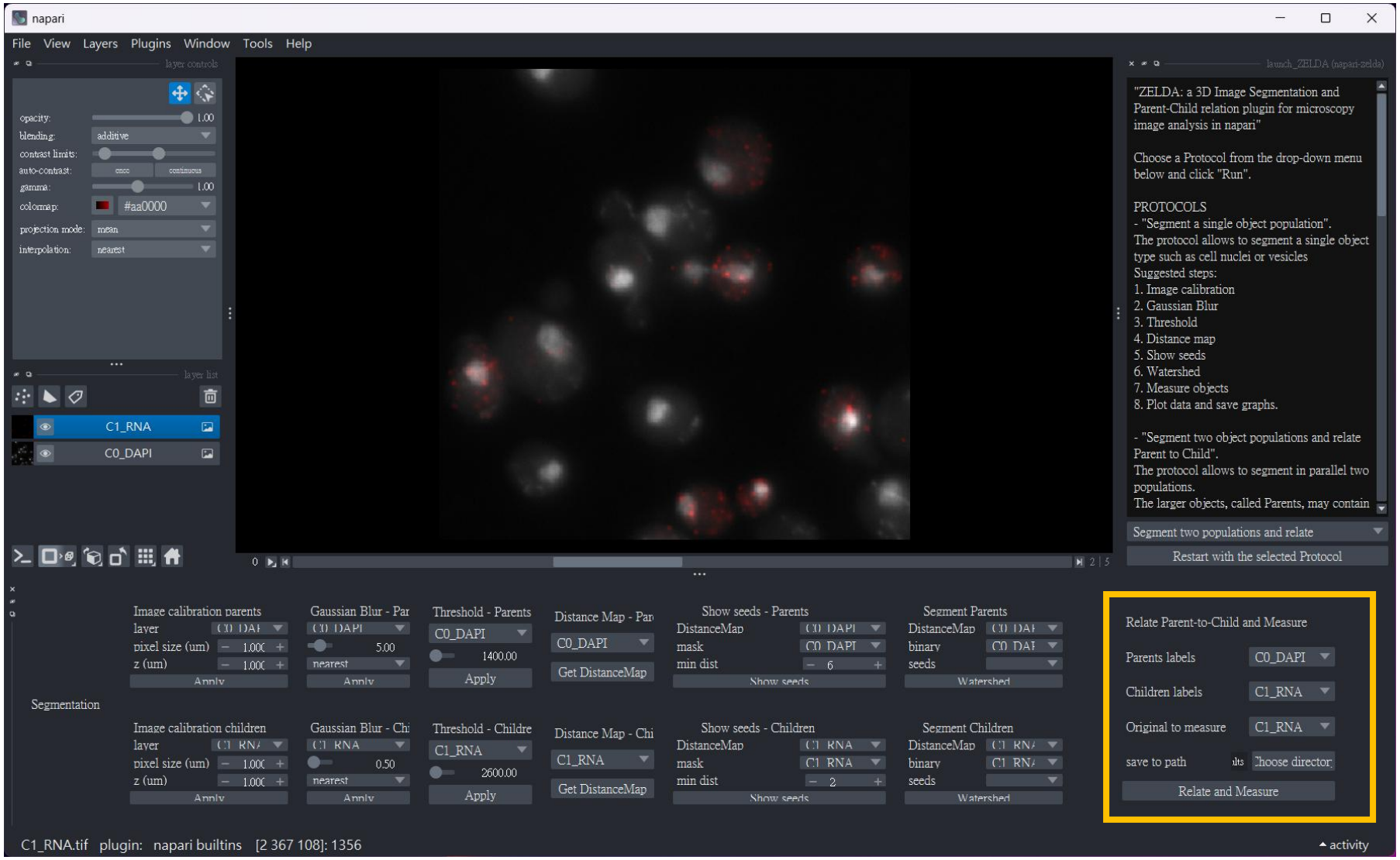
DAPI -> GB: sigma=**5.0**-> Th_parents=**1400.0**-> DistMap, Maxima: **min_dist=6**

RNA -> GB: sigma=**0.5**-> Th_children=**2600.0**-> DistMap, Maxima: **min_dist=2**



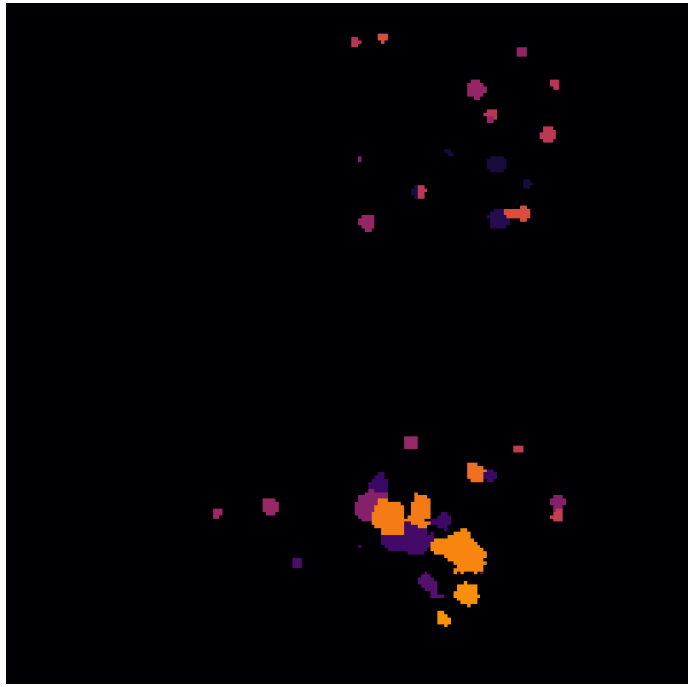
https://idr.openmicroscopy.org/webclient/?show=image-4496770

Segment two populations and relate parent-child relation

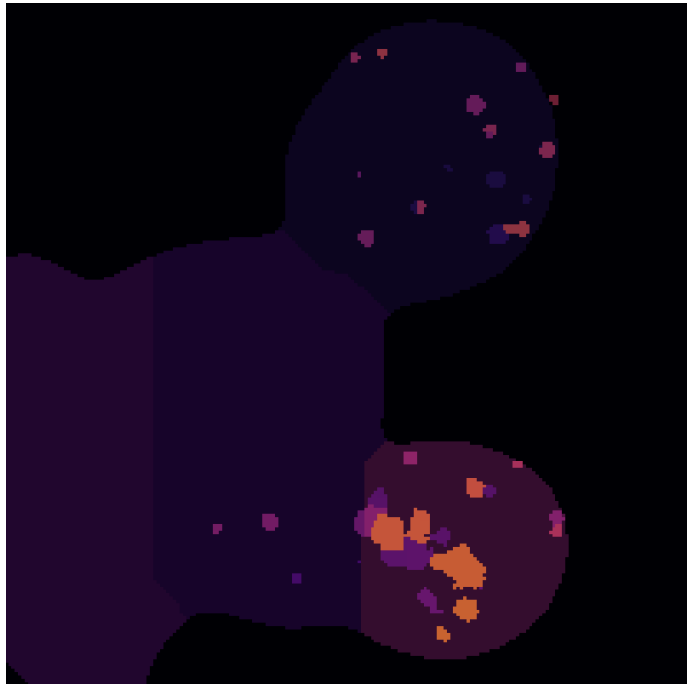


Labelled children objects by parents

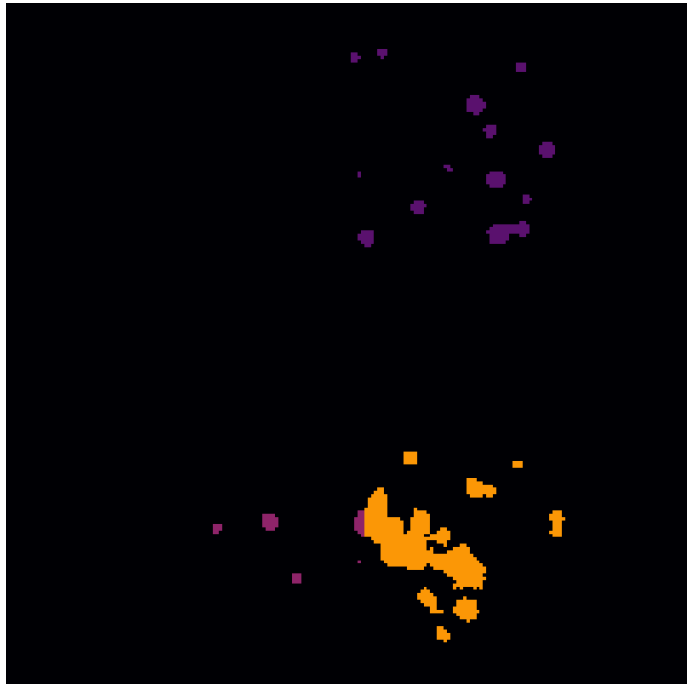
children_labels



parents_labels

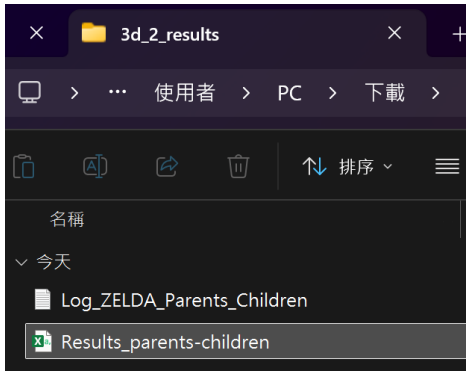


Find maximun of
parent_labels
for each child object



Results

Txt file: parameters for segmentation

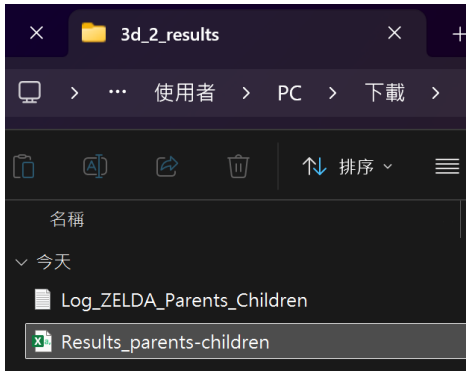


Csv file: results for parents and children objects (skimage.measure.regionprops)

1		Parent_label	Label	Area	Volume	Equivalent	MFI	Min_Intens	Max_Inten	MajorAxis	MinorAxis	Extent	Solidity	Centroid_X	Centroid_Y	Centroid_Z	Weigl
51	0	2	50	54	27	4.689556	3371	2529	4697	7.6202	2.902342	0.45	0.84375	294.3704	127.8704	0.361111	294.
52	0	2	51	36	18	4.096704	3353	2591	4714	6.494279	2.176271	0.72	0.947368	298.75	128.3889	0.222222	298
53	0	7	52	226	113	7.557357	3094	2508	4492	10.73732	5.705427	0.565	0.846442	466.2257	188.6858	0.962389	466.
54	0	0	53	39	19.5	4.207479	3108	2609	5079	5.058738	3.244012	0.52	0.928571	444.7692	191.1282	0.448718	444.
55	0	7	54	241	120.5	7.720987	3351	2492	4833	11.80199	5.486582	0.371914	0.782468	472.2946	192.7095	0.794606	472.
56	0	7	55	274	137	8.058434	3244	2513	5160	12.17874	4.713284	0.32426	0.756906	477.6679	197.9234	0.642336	477.
57	0	4	56	44	22	4.380107	3406	2577	4457	5.979399	2.430862	0.488889	0.897959	212.1364	198.7727	0.25	212.

Results

Txt file: parameters for segmentation



等效直徑

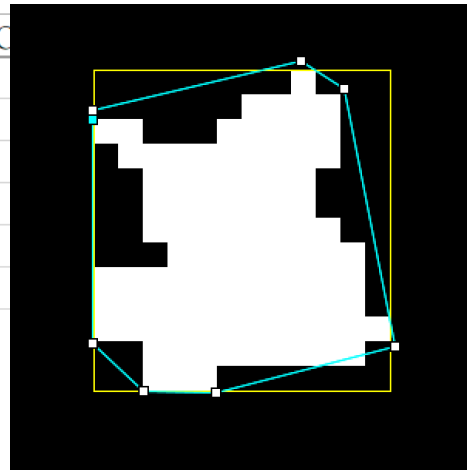
填充比例

實心度

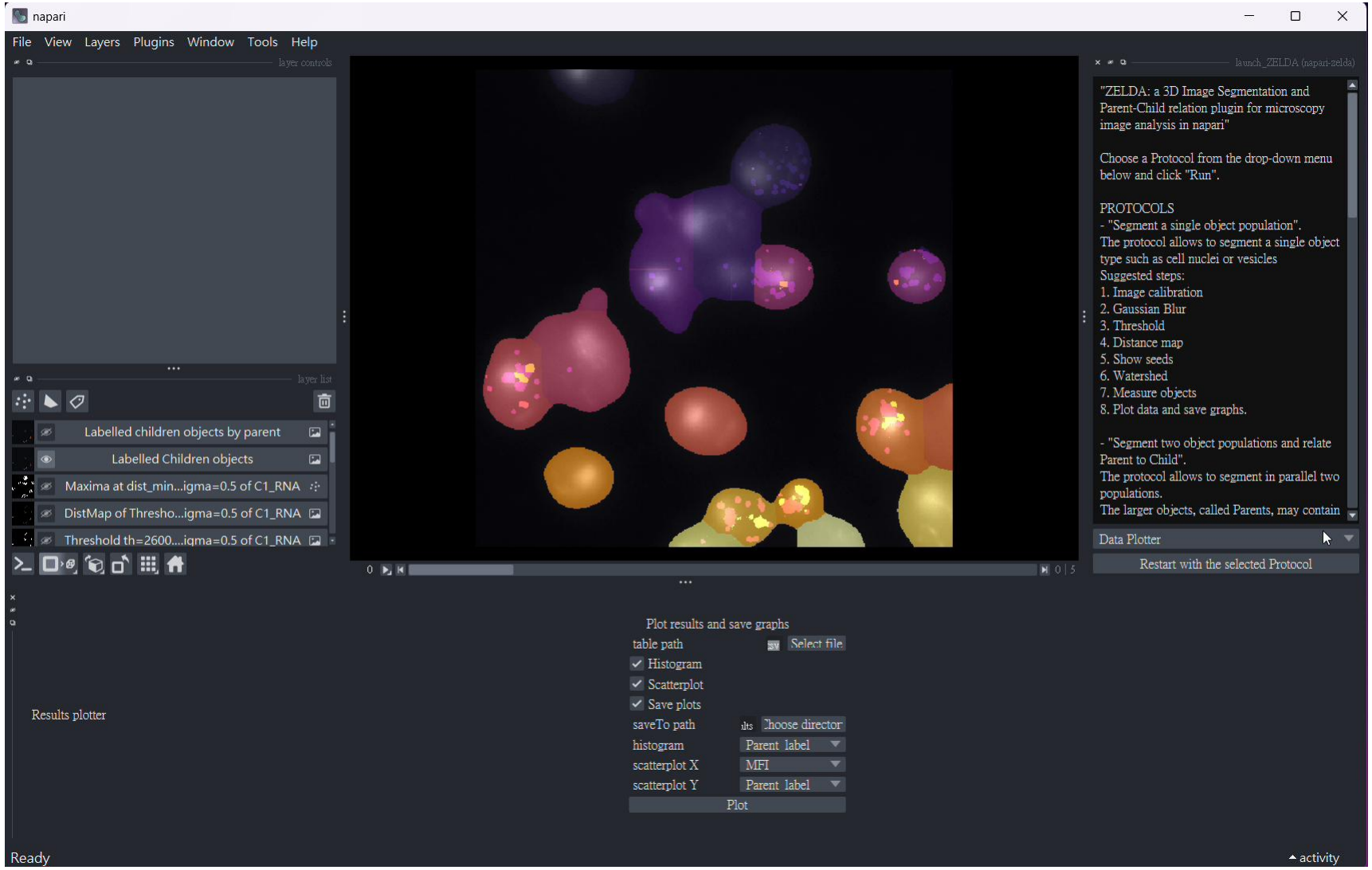
Csv file: results for parents and children objects

1		Parent_label	Label	Area	Volume	Equivalent	MFI	Min_Intens	Max_Inten	MajorAxis	MinorAxis	Extent	Solidity	C
51	0	2	50	54	27	4.689556	3371	2529	4697	7.6202	2.902342	0.45	0.84375	
52	0	2	51	36	18	4.096704	3353	2591	4714	6.494279	2.176271	0.72	0.947368	
53	0	7	52	226	113	7.557357	3094	2508	4492	10.73732	5.705427	0.565	0.846442	
54	0	0	53	39	19.5	4.207479	3108	2609	5079	5.058738	3.244012	0.52	0.928571	
55	0	7	54	241	120.5	7.720987	3351	2492	4833	11.80199	5.486582	0.371914	0.782468	
56	0	7	55	274	137	8.058434	3244	2513	5160	12.17874	4.713284	0.32426	0.756906	
57	0	4	56	44							62	0.488889	0.897959	

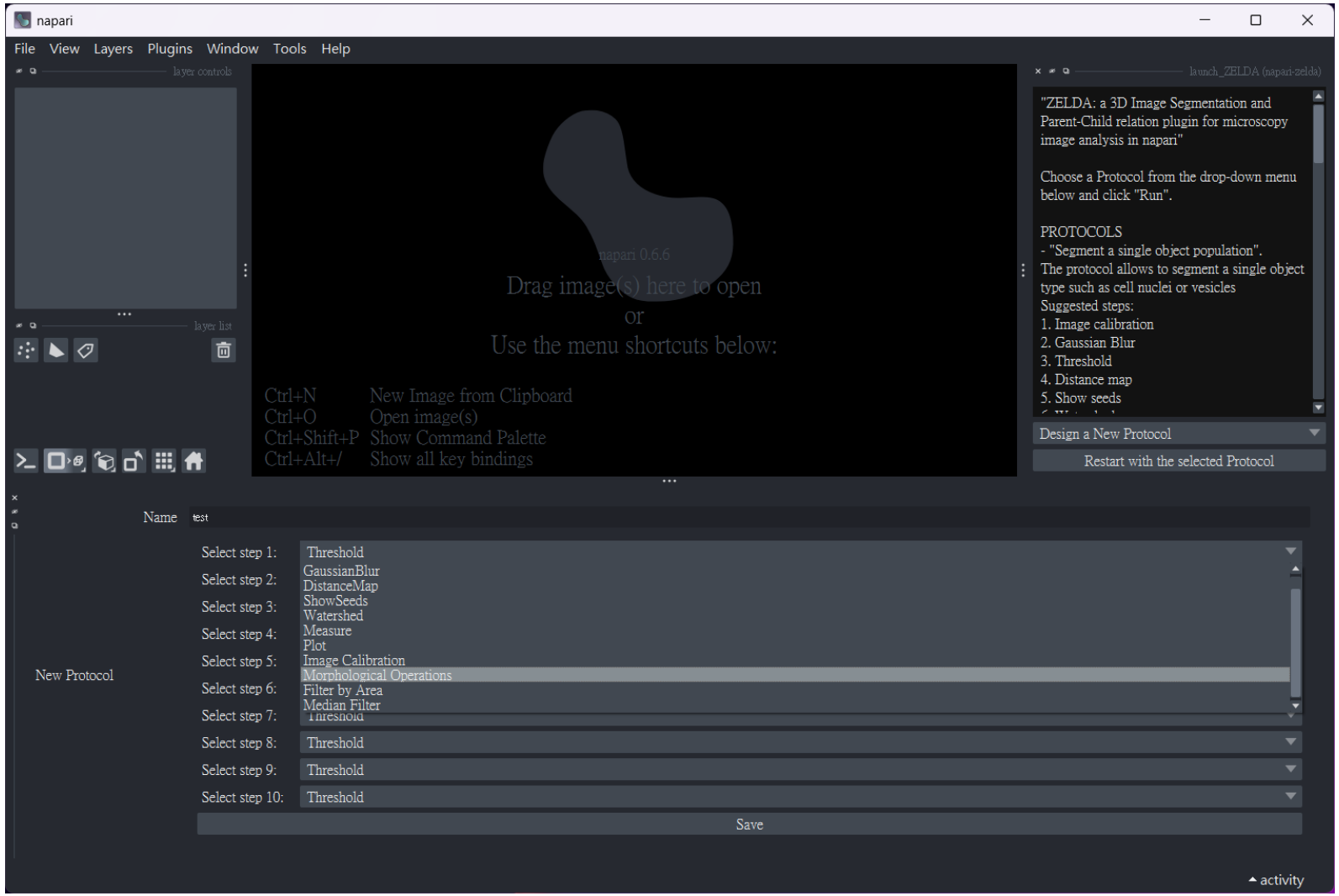
Mean Fluorescence Intensity
（平均螢光強度）



Zelda protocol: Data Plotter

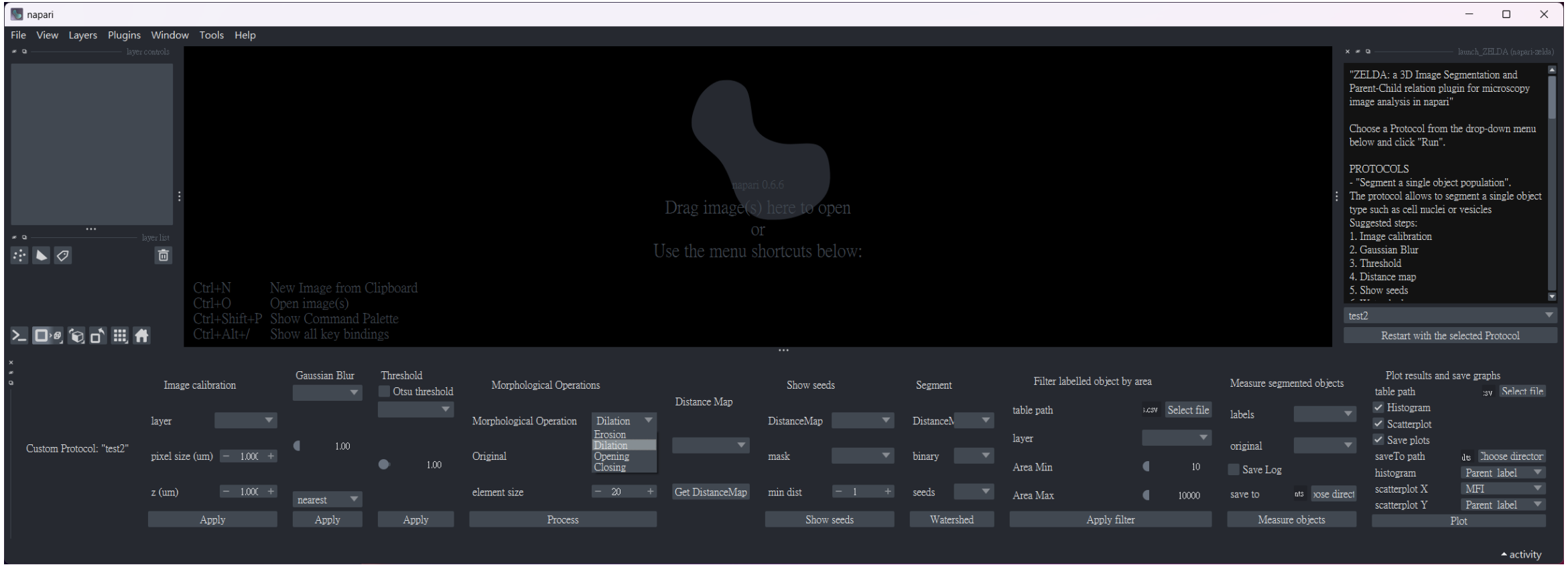


Zelda protocols: Design a New Protocol



Save > restart napari > Zelda > find the new ‘test’ protocols

Zelda protocols: Morphological operation, Filter labelled object by area, and Measure



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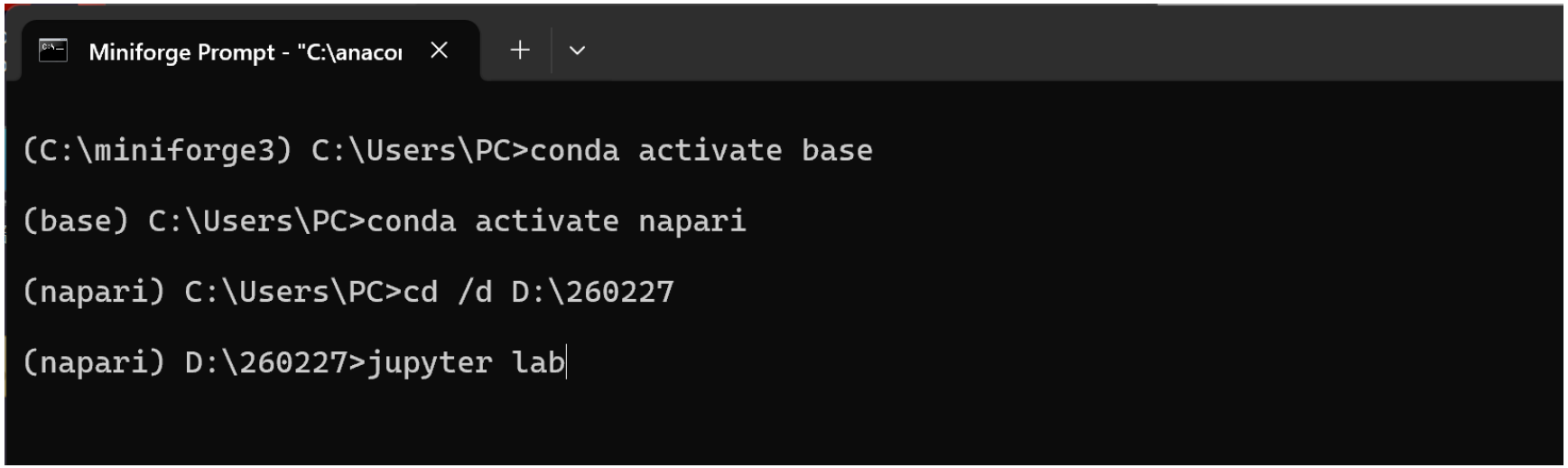
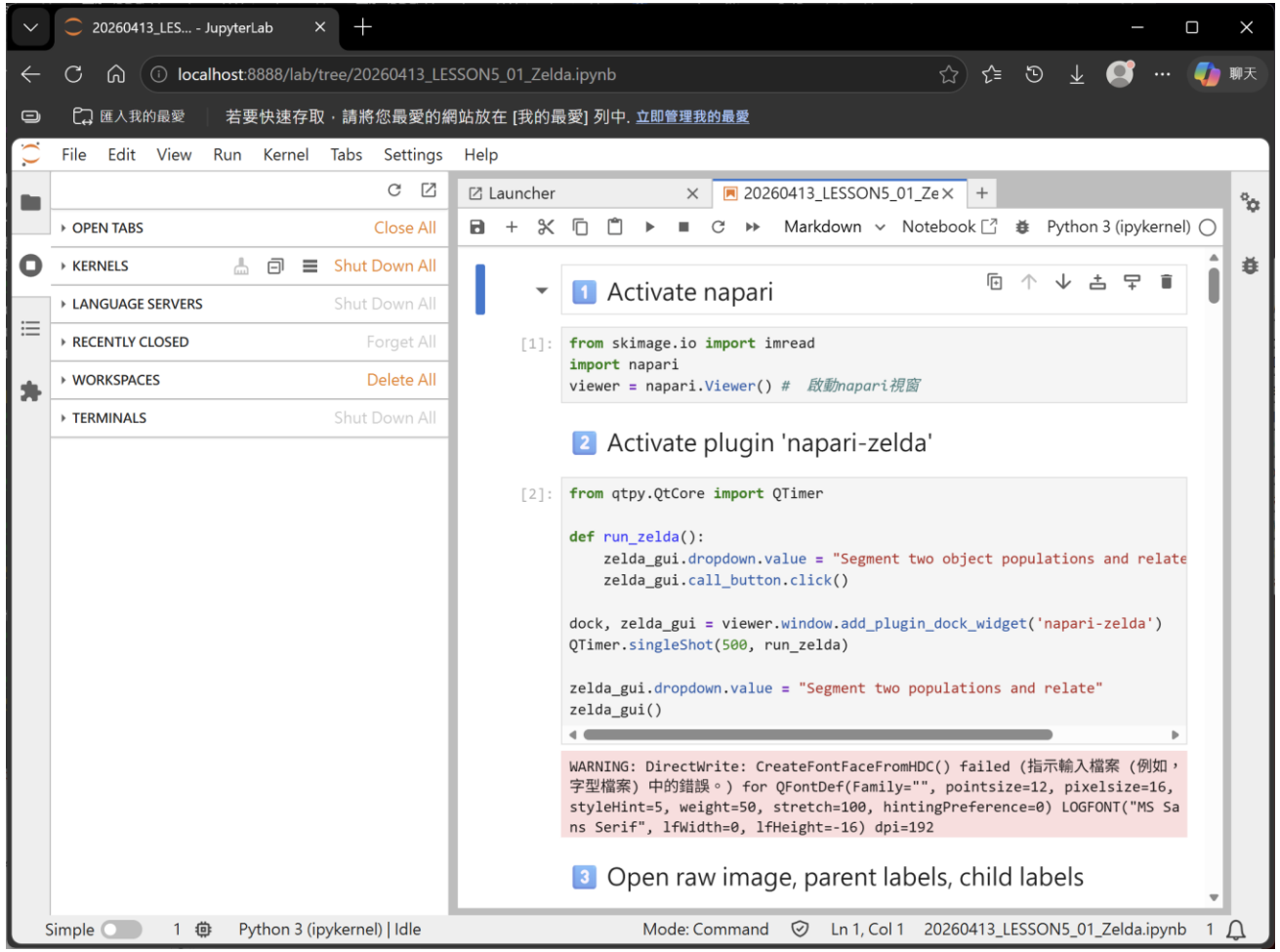
  - 使用Jupyter notebook 控制 napari-zelda
- Jupyter notebook batch 執行 parents-children relation analysis
  - 使用Jupyter notebook ，達到batch 執行parents-children relation analysis

使用Jupyter notebook 控制 napari-zelda

- Prompt conda activate 你的napari環境
  切換到 “20260413_LESSON5_01_Zelda.ipynb” 所在資料夾，
  啟動 jupyter lab

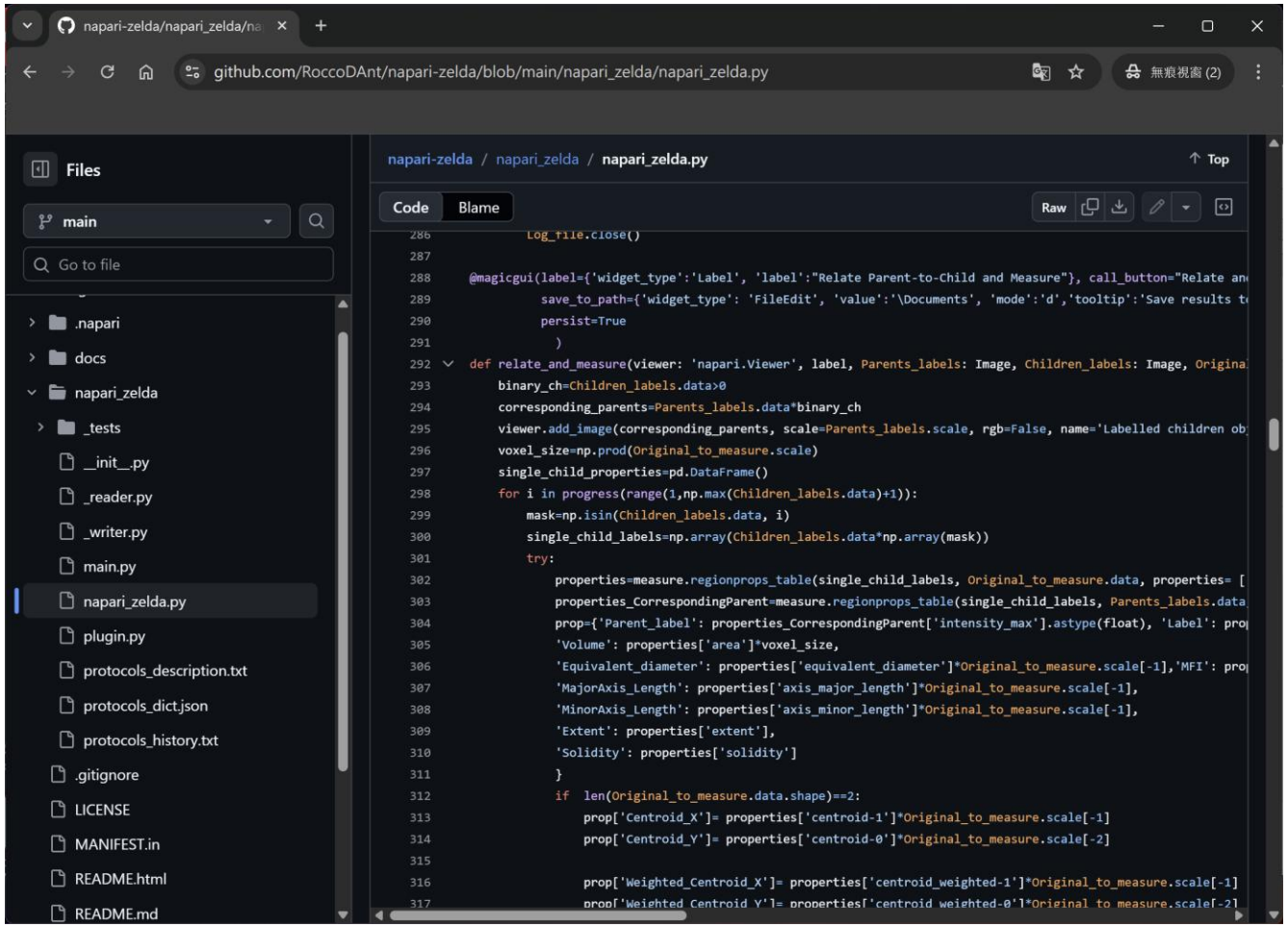


使用Jupyter notebook 控制 napari-zelda [20260413_Lesson05_Zelda.ipynb]



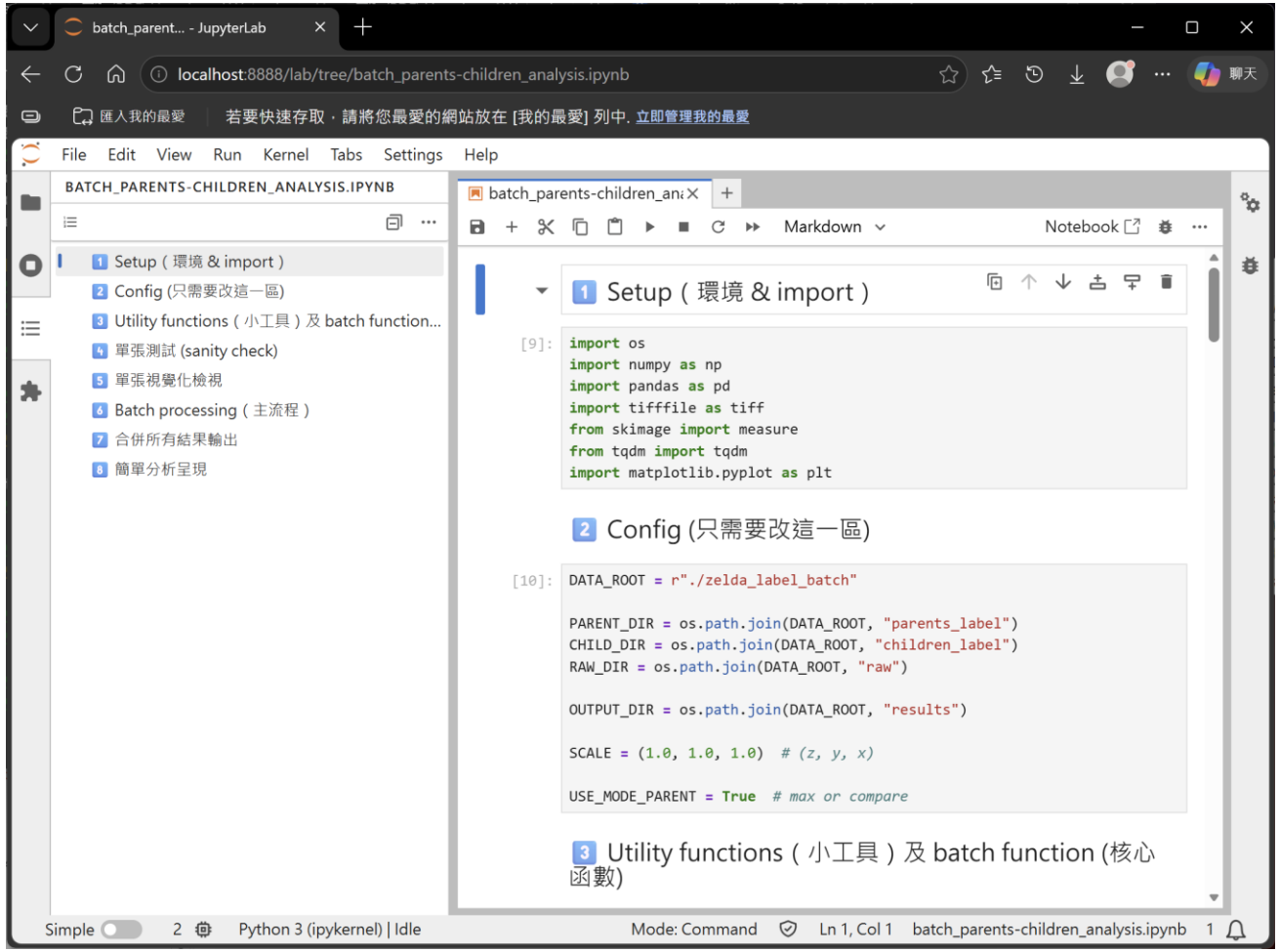
使用Jupyter notebook Jupyter notebook batch 執行 parents-children relation analysis

https://github.com/RoccoDAnt/napari-zelda



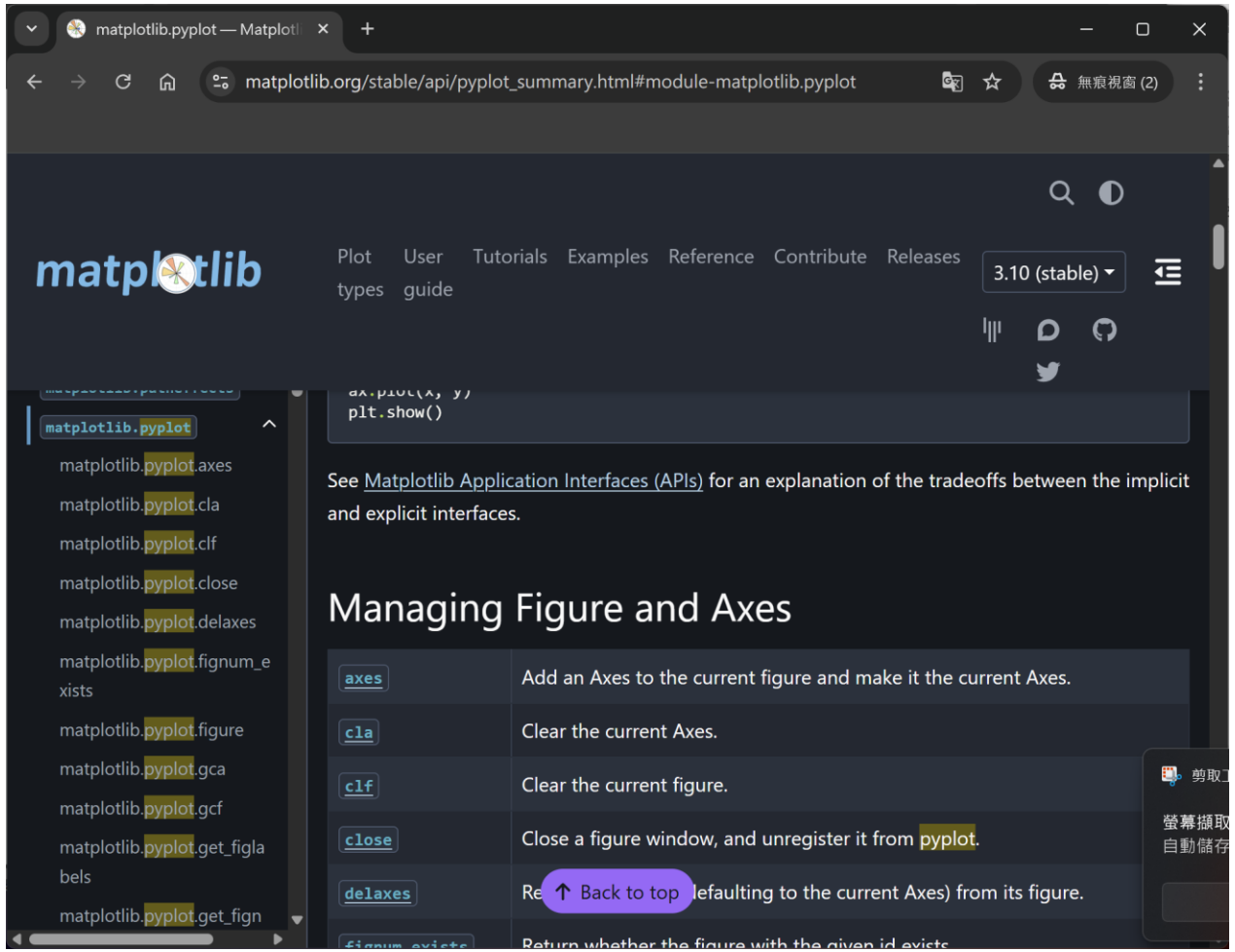
參考Github上 Zelda 原始程式碼，可以寫出 batch 執行 parents-children relation analysis 的 Jupyter notebook

Jupyter notebook batch 執行 parents-children relation analysis [batch_parents-children_analysis.ipynb]

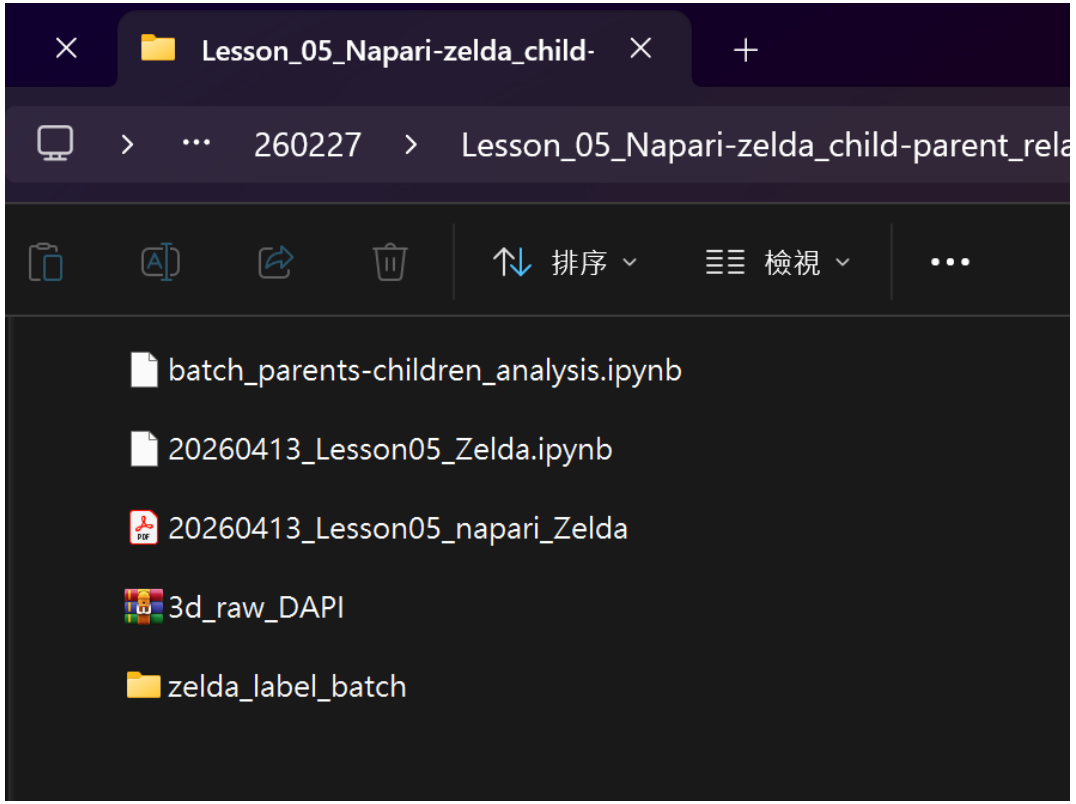


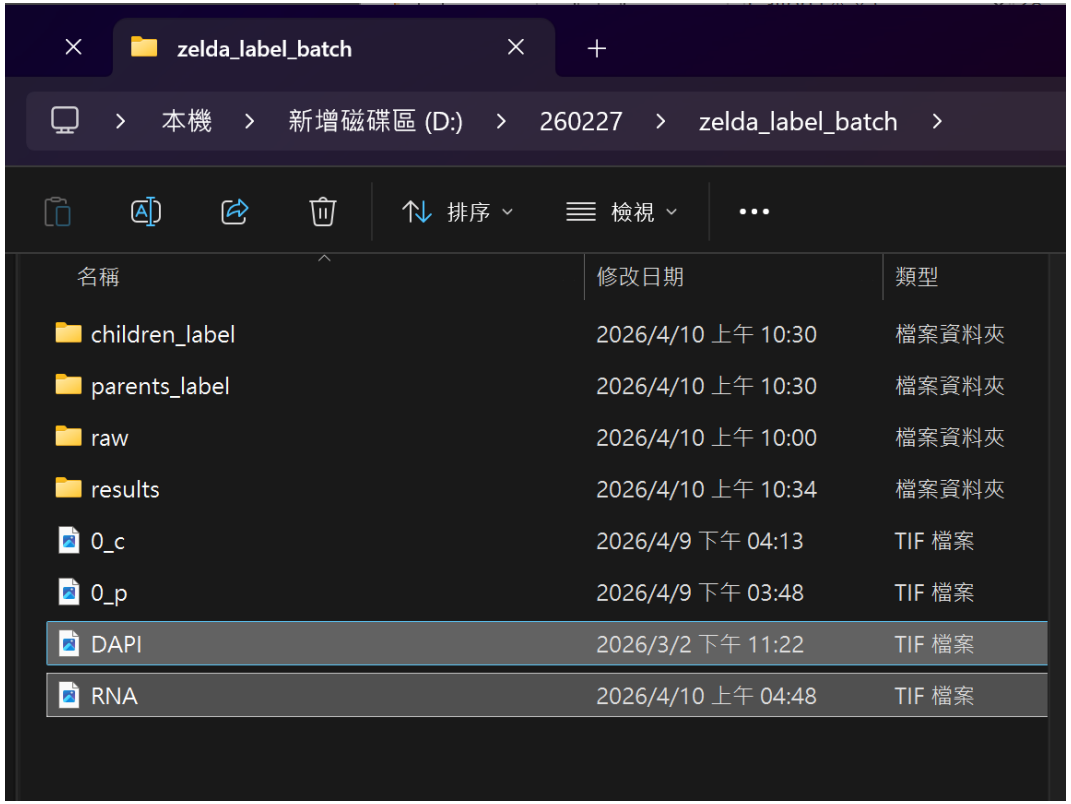
Matplotlib

https://matplotlib.org/stable/index.html



Sample image
zelda_labe_batch





Acknowledgements

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Thank you for your attention!